

☀ All Theory Questions of Chi-Square, ANOVA, Experimental Designs (CRD, RCBD, Latin Square), Regression & Multiple Regression (Including R^2) ☀

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📖 Source: Sher Muhammad Chaudhary (Theory Focused)

📝 Best for Mid/Final Exam Preparation

✅ 1. Chi-Square Test (χ^2):

1. What is the Chi-square test? State its uses and assumptions.
2. Explain the Chi-square goodness of fit test with an example.
3. What is the test of independence in Chi-square? How is it applied?
4. Describe the properties of the Chi-square distribution.
5. What is a contingency table? How is it used in Chi-square testing?
6. How are expected frequencies calculated in a Chi-square test?

✅ 2. Analysis of Variance (ANOVA):

1. Define ANOVA. What is its purpose in statistics?
2. What are the differences between one-way and two-way ANOVA?
3. Explain the concept of F-ratio and how it is interpreted.
4. What are the assumptions of ANOVA?
5. Describe the components of the ANOVA table (SS, MS, F, df).

✅ 3. Experimental Designs:

(A) Completely Randomized Design (CRD):

1. What is a Completely Randomized Design (CRD)? Explain its layout and assumptions.
2. How is the ANOVA table constructed for a CRD?
3. Discuss the advantages and limitations of CRD.

(B) Randomized Complete Block Design (RCBD):

1. What is a Randomized Complete Block Design? Explain with layout.
2. What is the purpose of blocking in an experiment?
3. How is the ANOVA table constructed in RCBD?
4. What are the merits and demerits of RCBD?

(C) Latin Square Design (LSD):

1. What is a Latin Square Design? In which situations is it used?
2. Explain the assumptions and layout of Latin Square Design.
3. How is the ANOVA table constructed for a Latin Square Design?

4. Regression and Multiple Regression:

Simple Linear Regression:

1. What is regression analysis? What is its importance?
2. Define the equation of a simple linear regression line.

3. Explain the least squares method used in regression.
4. What is the difference between regression and correlation?

Multiple Regression:

1. Define the multiple linear regression model.
2. Interpret the coefficients of a multiple regression equation.
3. What are the assumptions of multiple regression?
4. What is multicollinearity in regression analysis?

5. Coefficient of Determination (R^2):

1. What is the coefficient of determination (R^2)? What does it indicate?
2. What is the range and interpretation of R^2 ?
3. What is the difference between R^2 and adjusted R^2 ?

 **Best of Luck!** 

"Success is the sum of small efforts, repeated day in and day out." – Robert Collier

 Stay consistent, stay confident – your hard work will pay off!